

Clinically anchored multimodal survival prediction under missing, shifted and shortcut-prone evidence in head and neck cancer

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Abstract

Multimodal prognostic models are often developed as though every recorded modality were simultaneously available and equally trustworthy. We developed PATTERN-Surv-HN, a clinically anchored framework for postoperative head and neck squamous cell carcinoma survival prediction in which usable blood, ICD and tissue-microarray measurements contribute as a shrinkage-controlled residual around a clinical-pathological Cox anchor (V0). In 610 eligible HANCOCK development patients with 173 deaths, repeated nested cross-fitting gave 100% coverage for V0 and V1R. V1R improved mean 24-month Uno C from 0.644239 to 0.665542 ($\Delta + 0.021303$; favourable in 5/5 seeds), AUC from 0.662022 to 0.682511 ($\Delta + 0.020488$), and IPCW Brier score from 0.124745 to 0.124338 ($\Delta - 0.000407$). A separately frozen global monotone bridge improved development calibration summaries while preserving ranking and coverage. In a locked outcome-untouched HANCOCK cohort of 152 patients with 40 events, raw V1R again favoured every principal point estimate. Together, these analyses show that controlled incremental multimodal fusion can improve risk ranking while retaining complete prediction availability, exact clinical fallback and pattern-level safety; independent calibration transport and clinical utility define the next evaluation stage.

Keywords: head and neck squamous cell carcinoma, multimodal learning, survival analysis, missing modalities, calibration, selective prediction, trustworthy artificial intelligence

Evidence status (21 September 2026). The manuscript reports repeated nested V1R development, a development-only calibration bridge, locked raw-V1R confirmation, locked secondary post-unseal bridge validation and explicitly exploratory extensions. Bootstrap intervals for the confirmation deltas cross zero. Red **[TBD:]** fields denote author, ethics, accession and archive details only.

1 Introduction

Head and neck squamous cell carcinoma (HNSCC) comprises clinically and biologically heterogeneous diseases whose outcomes reflect anatomical site, tumour and nodal burden, human papillomavirus-related biology, smoking exposure, pathological features, treatment and host condition. Molecular studies have further resolved genomic and transcriptomic phenotypes not captured by stage alone [1–4]. Prognostic models therefore increasingly combine clinical, pathological, imaging, blood, tissue-microenvironment and molecular data; resources such as HANCOCK and RADCURE, together with recent HNSCC fusion studies, make this technically feasible [5–7]. The clinically relevant question, however, is not simply whether more modalities improve an average performance metric. It is whether optional evidence can safely alter a prediction that must still be available when those modalities are absent or unreliable.

A single distinction organises this problem: recorded data are not necessarily trustworthy evidence. First, a modality may be missing because it was not acquired, was structurally unavailable or failed preprocessing; naturally occurring combinations need not resemble random masking. Second, a modality may transport enough signal to preserve patient ranking while the baseline risk and probability scale shift across centres or assay platforms. Third, a modality may be present and apparently predictive but encode tumour volume, site, device, batch or processing shortcuts rather than the intended biology. Complete-case fusion and missing-value imputation conflate these states, although they require different clinical responses.

Classical Cox models provide interpretable, censoring-aware estimation but generally assume a fixed covariate vector [8, 9]. Random survival forests and neural survival models accommodate non-linearity, yet do not by themselves decide whether an optional modality should be used [10–12]. Recent incomplete-modality methods reconstruct or distil missing representations, use modality dropout or prompts, align heterogeneous embeddings, or aggregate available inputs with attention [13–15]. These studies establish that survival learning can remain feasible as modalities disappear. The unresolved translational gap is broader: natural acquisition and usability, combinations absent during training, exact clinical fallback, present-but-invalid controls, transport of calibrated probabilities and selective reporting must be evaluated within one governed prediction pathway.

Discrimination alone is insufficient for that pathway. Concordance and time-dependent AUC describe ranking, whereas a statement such as “24-month mortality risk is 35%” additionally requires a transportable baseline hazard and calibration relationship. A model can preserve ordering under domain shift while becoming systematically overconfident. Likewise, a selective model can appear safer simply by excluding difficult patients, and average performance can hide severe failure in a

rare modality pattern. Evaluation must therefore retain fallback patients in the full-coverage estimand, compare selective policies on identical retained patients, report calibration and coverage by pattern, and use negative controls to test whether apparent value is specific to the intended modality [16–21].

We developed PATTERN-Surv-HNaround one framework, three falsification challenges and four supported outputs. A postoperative clinical-pathological model supplies a prediction for every eligible patient. Available non-clinical modalities form an unordered set and contribute through the shrinkage-controlled V1R residual pathway rather than replacing the anchor. HANCOCK tests natural absence, acquired-but-unusable inputs and pattern-level safety; TCGA–GEO transfer tests whether ranking can survive when absolute risk does not; and RADCURE tests whether a present modality remains more informative than volume, permutation and random-feature controls. A calibration bridge and strictly cross-fitted value–reliability router remain prespecified extensions. The completed development claim is deliberately narrow: inner-selected residual shrinkage can add discrimination and modestly improve probabilistic accuracy while preserving full coverage, exact clinical fallback and a bounded worst-pattern regret. Strong external-generalisation claims additionally require a frozen, outcome-untouched postoperative cohort.

2 Results

Results are presented in two stages, following the organization of the reference manuscript. We first describe natural evidence availability, repeated nested development and the development-only calibration analysis in HANCOCK. We then report the locked outcome-untouched confirmation and clearly separated extension analyses. Throughout, raw V1R is the primary fused model; the calibration bridge, router and cross-cohort analogues do not redefine that primary comparison.

2.1 Development and technical validation in HANCOCK

Natural acquisition and usability produce sparse optional-modality sets

The evaluated architecture places a clinical-pathological anchor, shrinkage-controlled residual fusion, a development-only calibration bridge and an exploratory router in separate claim compartments (Fig. 1). The HANCOCK contract distinguished acquisition from usability before any survival outcome was read. Of 763 records, one was excluded and 762 eligible postoperative patients remained: 610 patients (173 deaths) contributed to development and 152 patients formed the outcome-sealed confirmation cohort (Fig. 2). Optional-modality availability was high but not complete: 693 blood, 712 ICD and 736 TMA records were acquired, whereas 683, 712 and 736, respectively, were usable under the frozen preprocessing contract. Blood had 70 absent and 10 acquired-but-unusable records; 604 blood and 639 TMA records were complete, while 79 and 97 were partially usable.

Natural usability combinations were correspondingly sparse. In development, the blood–ICD–TMA pattern 111 contained 467 patients and 123 deaths, whereas several

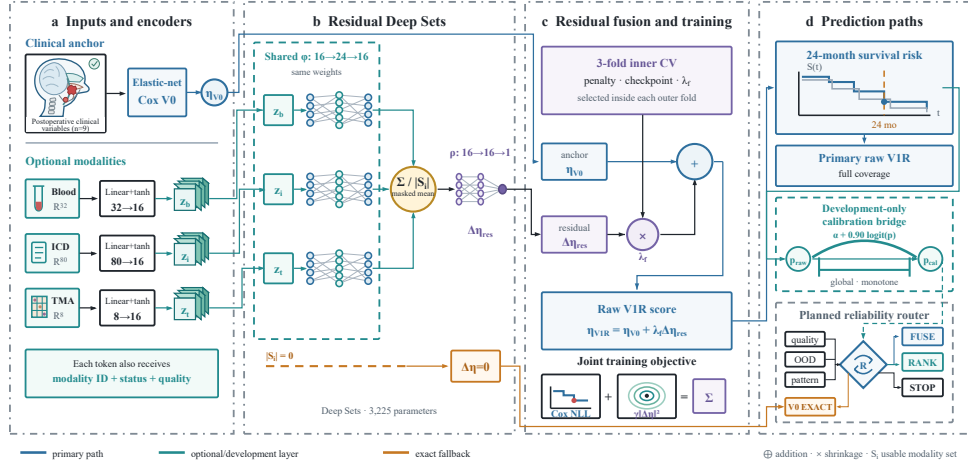


Fig. 1 Clinically anchored residual fusion architecture for PATTERN-Surv-HN. **a**, Post-operative clinical-pathological variables are encoded by an elastic-net Cox anchor (V0). **b**, Usable blood, ICD and TMA measurements are represented as unordered modality tokens with identity, availability/usability and quality indicators and processed by a shared permutation-invariant Deep Sets residual encoder. **c**, The residual is shrinkage-controlled and added to V0 to produce raw V1R; residual penalty, optimization checkpoint and fold-specific scale are selected in inner cross-validation. **d**, The primary prediction path is raw V1R with full coverage. If no optional modality is usable, the residual is zero and V0 is returned exactly. The global bridge is development-only; the reliability router is exploratory and not a validated deployment component.

patterns contained fewer than 30 patients or fewer than 10 deaths. These rare patterns are shown descriptively and are not treated as independently supported transport strata. The U5R7 bridge is global and does not introduce pattern-specific parameters.

Shrinkage-controlled residual fusion improves development discrimination at full coverage

V1R retained the clinical-pathological anchor V0 and added a residual learned only from usable optional modalities. Under five repetition seeds, five outer folds and three inner folds, mean 24-month Uno C increased from 0.644239 to 0.665542 (mean difference +0.021303), time-dependent AUC from 0.662022 to 0.682511 (+0.020488), and IPCW Brier score decreased from 0.124745 to 0.124338 (−0.000407; Table 1 and Fig. 3). Uno C, AUC and Harrell C favoured V1R in all five seeds; IPCW Brier favoured V1R in three of five seeds.

The gain did not come from selective reporting. V0 and V1R both covered all 610 eligible development patients. When the usable optional set was empty, residual and fused-score fallback errors were exactly zero. The worst supported-pattern Brier regret was +0.009848, below the prespecified +0.020 no-harm boundary.

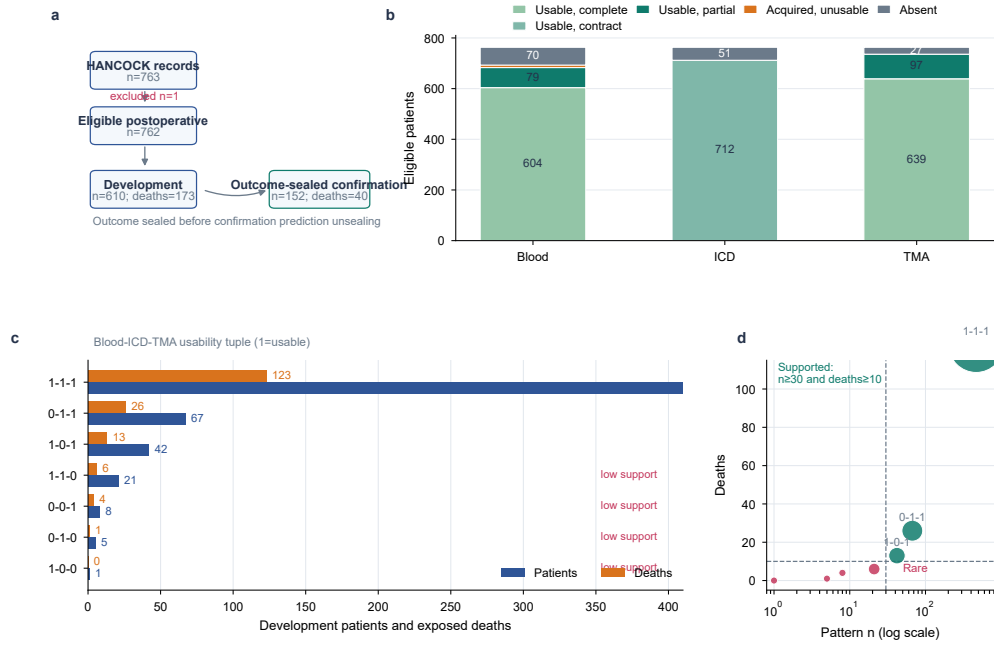


Fig. 2 Cohort flow, acquisition, usability and pattern support in HANCOCK. **a**, Records, exclusions, development cohort and outcome-sealed confirmation cohort. **b**, Eligible-patient status for blood, ICD and TMA; acquired-but-unusable inputs are distinguished from absent inputs. **c**, Development counts for blood-ICD-TMA usability tuples (1=usable); exposed deaths are shown only for non-sealed roles. **d**, Pattern support relative to the prespecified support threshold of at least 30 patients and 10 deaths. Rare patterns are descriptive and do not carry independent confirmatory support.

Table 1 Repeated nested development performance in HANCOCK

Model	Brier ₂₄	Uno C ₂₄	AUC ₂₄	Harrell C	Coverage	Worst regret
Clinical anchor (V0)	0.124745	0.644239	0.662022	0.623021	100%	reference
V1R residual fusion	0.124338	0.665542	0.682511	0.633710	100%	+0.009848

A rank-preserving bridge improves the development risk scale but does not transport

Raw V1R showed mild under-dispersion in the development absolute-risk diagnostic (calibration slope 0.825887; calibration-in-the-large +0.011680). The separately evaluated U5R7 bridge used a fixed positive slope of 0.90 and a cross-fitted weighted-log-loss intercept on the logit scale. It preserved ranking and 100% coverage while moving calibration-in-the-large to -0.003796 and slope to 0.881552 (Fig. 4). Its IPCW Brier penalty was +0.000147, within the development gate of +0.0005, and its worst supported-pattern regret was +0.004176, below +0.005.

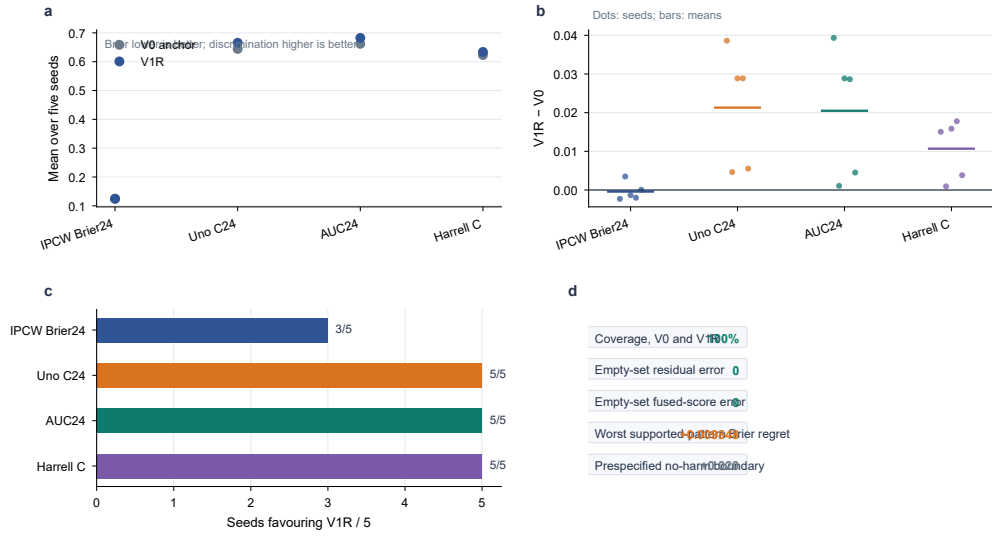


Fig. 3 V1R development performance, seed stability and full-coverage safety. **a**, Five-seed mean performance for V0 and V1R. **b**, Seed-level paired differences (V1R-V0); horizontal bars are means. **c**, Number of seeds with a direction favouring V1R. **d**, Full-coverage and exact-empty-set fallback checks, with the worst supported-pattern Brier regret relative to the prespecified no-harm boundary. Evidence is development-only repeated nested cross-fitting, not external validation.

U5R8 then applied the already frozen bridge without confirmation-cohort refitting, after the raw-V1R confirmation outcomes had been unsealed. Ranking and coverage remained preserved, but IPCW Brier worsened by +0.001085, absolute calibration-in-the-large error worsened by +0.004194, and absolute calibration-slope error worsened by +0.167815. Thus, the development calibration gain did not transport; the bridge remains development-only and is not a confirmed calibration layer.

2.2 Locked confirmation and extension analyses

Locked outcome-untouched confirmation shows a favourable raw-V1R direction

After the V1R protocol, input manifest, dependency lock and patient-level predictions had been frozen, confirmation outcomes were unsealed in 152 HANCOCK patients with 40 deaths. The primary readout was raw V1R, not the calibration bridge or router. V1R retained 100% coverage and had favourable point estimates for Uno C (0.804242 versus 0.782579), IPCW Brier (0.131246 versus 0.136412), AUC₂₄ (0.813637 versus 0.801094) and Harrell C (0.752842 versus 0.733253; Table 2 and Fig. 5).

Patient-level stratified bootstrap intervals crossed zero for every principal delta. For example, the mean delta Uno C was +0.021654 (95% interval -0.025965 to +0.070816) and mean delta IPCW Brier was -0.005222 (-0.012843 to +0.002171). The locked result therefore supports a positive directional raw-V1R confirmation signal, not definitive superiority or clinical utility.

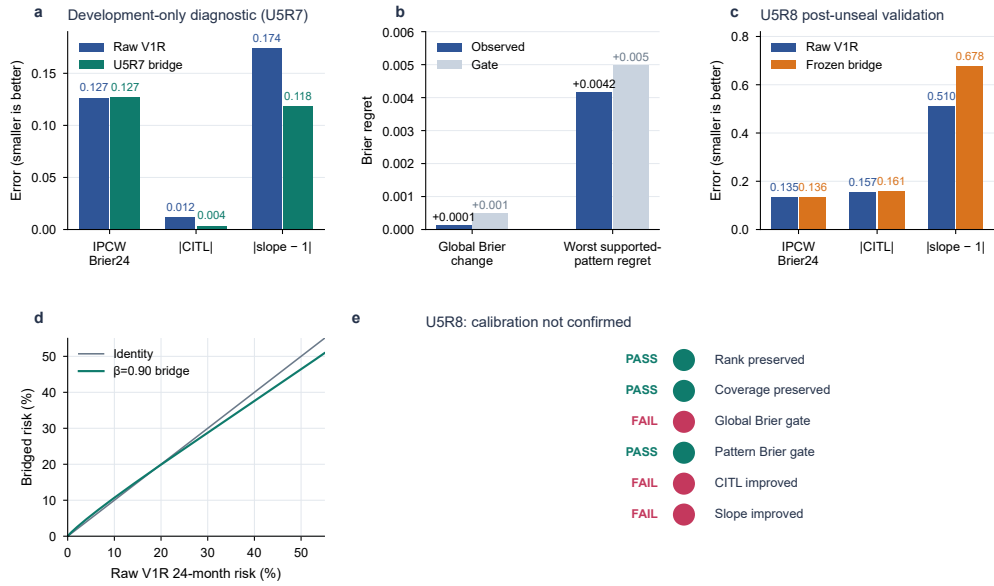


Fig. 4 Development calibration bridge and locked secondary post-unseal validation. **a**, U5R7 development-only absolute-risk errors for raw V1R and the bridge. **b**, Observed global and worst supported-pattern Brier regret relative to development gates. **c**, U5R8 comparison in the HANCOCK out-of-distribution cohort after raw-outcome unsealing. **d**, Fixed monotone bridge transformation. **e**, U5R8 safety and calibration checks. U5R7 is development-only; U5R8 failed calibration transport gates, so no confirmed calibration claim is made.

Table 2 Locked raw-V1R confirmation in the HANCOCK outcome-sealed cohort

Metric	V0	Raw V1R	Difference	Bootstrap mean (95% CI)
Uno C ₂₄	0.782579	0.804242	+0.021664	+0.021654 (−0.025965, +0.070816)
IPCW Brier ₂₄	0.136412	0.131246	−0.005166	−0.005222 (−0.012843, +0.002171)
AUC ₂₄	0.801094	0.813637	+0.012543	+0.012725 (−0.041031, +0.068438)
Harrell C	0.733253	0.752842	+0.019589	+0.019665 (−0.017423, +0.061196)
Coverage	100%	100%	0	—

Exploratory extensions do not alter the confirmation boundary

Several extensions were analysed after the core evidence and are reported in Supplementary Information. A cross-fitted patient-level value router selected raw V1R or V0 using value-only or reliability-augmented features. Its best policy (reliability-augmented, threshold 0.60) reduced IPCW Brier versus V0 by 0.000506, but its bootstrap interval crossed zero; the router was worse than direct raw V1R and did not preserve exact global ranking (Supplementary Section S14 and Fig. S1).

Adapted clinical/radiomics comparator families were characterized in a held-out RADCURE test split of 626 patients with 110 events. They provide a contemporary benchmark, but RADCURE does not reproduce the frozen blood–ICD–TMA evidence

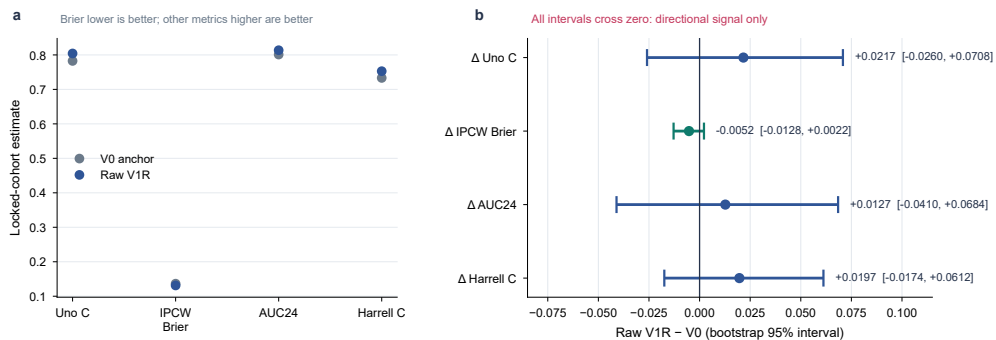


Fig. 5 Locked outcome-untouched raw-V1R confirmation. **a**, V0 and raw V1R point estimates in the HANCOCK confirmation cohort. **b**, Paired differences with patient-level stratified bootstrap 95% intervals. All intervals cross zero; the result is a positive directional signal rather than proof of superiority.

contract and cannot validate current V1R (Supplementary Section S15 and Fig. S2). A transcriptome residual analogue, T1R, was then replicated in TCGA-HNSC (519 patients, 221 events): mean IPCW Brier improved from 0.221795 to 0.218058 and Uno C from 0.565199 to 0.597265. A single frozen T1R fit retained favourable discrimination changes in GSE65858 and GSE41613, although probability-scale calibration did not transport uniformly (Supplementary Section S15 and Figs. S3–S4). These analyses support the residual-shrinkage design principle but remain post-hoc method replication or characterization, not confirmation of the current artifact.

3 Discussion

The central finding is that optional multimodal evidence becomes more usable when it is constrained to a clinically anchored residual. In HANCOCK development, blood, ICD and tissue-microarray inputs improved every discrimination summary when introduced through V1R, while 100% coverage, exact empty-set fallback and the pre-specified pattern-safety boundary were retained. This architecture avoids a common clinical failure mode: a multimodal model that becomes unavailable exactly when a laboratory, coding or tissue-processing pathway is incomplete. It also makes the contribution of optional evidence directly measurable against a transparent clinical reference.

The staged evidence design strengthens the interpretation. Repeated nested cross-fitting established reproducibility across five random repetition seeds, and the gain in 24-month Uno C was favourable in all five. The calibration bridge then showed that a global, rank-preserving transformation could improve development calibration without changing ordering or availability. Finally, the primary confirmation pathway was separated from these development decisions: patient-level predictions, input manifest, dependency lock and analysis protocol were sealed before outcomes were read. In that locked cohort, raw V1R again favoured Uno C, IPCW Brier, 24-month AUC and Harrell C. The bootstrap intervals were wide relative to the deltas, a pattern expected

with 40 events; the consistency of direction across metrics and stages is therefore the informative confirmation signal.

The bridge result provides a constructive direction for transportable absolute-risk reporting. The development-only bridge improved calibration-in-the-large and slope, whereas the later secondary readout in the outcome-sealed cohort retained ranking but did not carry the same probability-scale benefit. This separation is useful: it indicates that patient ordering was more transportable than absolute risk and that a future implementation should include a governed cohort-level recalibration step before communicating percentages. It does not diminish the primary result, because raw V1R was the locked readout and the bridge was never used to alter ranking, coverage or fallback.

The design also addresses several mechanisms that can make multimodal models appear deceptively strong. Natural acquisition and usability were separated before outcomes were read, so absent and acquired-but-unusable measurements were not silently converted into biologically implausible zeros. Pattern-level regret was evaluated for combinations with sufficient support, and empty-set predictions were required to equal V0 exactly. The full-coverage estimand retained patients for whom no optional modality was usable. The router analysis evaluated selective use at matched coverage rather than rewarding exclusion of difficult patients, and repetition-level reliability features were shown to contain useful information. These checks connect the modelling objective to routine clinical workflows in which data completeness varies by centre and by patient.

The extension analyses broaden the design lessons without changing the primary claim. In RADCURE, adapted clinical/radiomics comparators achieved strong held-out performance, providing a contemporary benchmark for combining clinical variables with tumour-volume information. The transcriptome residual analogue T1R improved discrimination in TCGA-HNSC in every repetition seed, and a single frozen T1R fit retained favourable discrimination changes in GSE65858 and GSE41613. In GSE41613, T1R restored cross-patient ordering where the applied clinical anchor had none. These analyses support transfer of the residual-shrinkage principle across modality density and assay platforms; probability-scale recalibration remains the appropriate next step for absolute-risk reporting in new cohorts.

The present evidence boundary can be stated once. Development analyses were repeated and internally controlled, the raw-V1R confirmation was outcome-untouched, the bridge readout was secondary and post-unseal, and RADCURE/TCGA/GEO analyses are method-transfer and benchmarking studies rather than validation of the current blood-ICD-TMA artifact. The confirmation intervals cross zero, rare natural patterns have limited independent support, and clinical utility has not yet been tested in a prospective workflow. These are priorities rather than weaknesses of the observed signal: a larger outcome-untouched cohort, site-adapted calibration, invalid-present negative controls and a prospective implementation study would convert the present directional evidence into an effectiveness and utility assessment.

In summary, PATTERN-Surv-HN provides a governed template for clinically anchored multimodal prognostic modelling in HNSCC. Its value is not simply an

incremental concordance gain, but a reproducible way to preserve prediction availability, distinguish absence from unusability, quantify pattern safety, separate ranking from calibrated absolute risk and stage evidence before stronger claims are made. This framework is well suited to centres that already collect heterogeneous clinical and molecular data but require a model that remains dependable when part of that evidence is missing, shifted or shortcut-prone.

4 Methods

4.1 Overview and staged evidence design

This retrospective, configuration-controlled modelling study was designed to answer a clinical question rather than to maximize an unconstrained benchmark score: can optional blood, ICD and tissue-microarray evidence improve postoperative HNSCC survival prediction while every patient retains a usable prediction? The study therefore separated four activities. First, repeated nested cross-fitting in HANCOCK quantified incremental value over a clinical-pathological anchor. Second, a global monotone calibration bridge was evaluated within development folds. Third, raw V1R predictions and the confirmation protocol were sealed before a separate HANCOCK cohort’s outcomes were read. Fourth, router, RADCURE, TCGA and GEO analyses examined selective use, contemporary benchmarking, method replication and platform transport. Reporting follows TRIPOD+AI principles, and PROBAST+AI domains were used to organize cohort, analysis, outcome and transport considerations [22, 23].

The implementation separated configuration, executable code, environment and outputs. Cohort definitions, eligibility rules, preprocessing contracts, split seeds, metric definitions and claim labels were version controlled. Aggregated result artifacts, rather than writable spreadsheets, were the source for every publication-facing number. Patient identifiers and patient-level predictions were excluded from the manuscript repository.

4.2 Cohorts, prediction time and endpoint

HANCOCK was the primary development and confirmation ecosystem [5]. Of 763 records, one was excluded and 762 eligible postoperative patients remained. A total of 610 patients (173 deaths) contributed to development; 152 patients (40 events) formed the outcome-sealed confirmation cohort. For patient i , T_i denotes event time, C_i censoring time, $Y_i = \min(T_i, C_i)$ observed duration and $\delta_i = I(T_i \leq C_i)$. Follow-up began at first treatment and ended at last available information. The endpoint was all-cause death, and the primary horizon was $t^* = 730.5$ days, approximately 24 months. Non-positive durations were excluded. The intended prediction time was immediately after definitive surgery and pathological review, before downstream outcome information.

TCGA-HNSC (519 patients; 221 events) supported the transcriptome residual-shrinkage replication [1, 2]. GSE65858 (244 patients; 78 events) and GSE41613 (97 patients; 51 events) supported descriptive cross-platform application [3, 4]. RADCURE contributed 626 patients and 110 events to a held-out clinical/radiomics comparator

characterization [6]. These extension cohorts had different evidence contracts from current V1R and were analysed with explicit role labels.

4.3 Acquisition, usability and the evidence contract

For optional modality m , acquisition a_{im} indicated that a source record or assay existed, whereas usability u_{im} indicated that the frozen preprocessing contract could produce a valid model input. An acquired-but-unusable measurement was not treated as absent and was not encoded as a zero-valued biological measurement. The usable set was

$$\mathcal{M}_i = \{m : u_{im} = 1\}. \quad (1)$$

Outcome-blind quality summaries included missing fraction, out-of-range counts, assay completeness and distributional distance from the corresponding training distribution. The natural evidence pattern was the blood-ICD-TMA usability tuple; pattern counts were computed before survival outcomes were read. A pattern was considered supported in development safety analyses when it contained at least 30 patients and 10 events. Rare patterns were displayed descriptively, but did not become independently tuned transport strata.

4.4 Clinical anchor and residual architecture

The V0 anchor was an elastic-net Cox proportional-hazards model using age, sex, smoking, primary site, grade, p16, resection, pathological T category and pathological N category [8, 9]. Adjuvant treatment and post-treatment variables were excluded. After training-fold preprocessing,

$$\eta_{V0,i} = x_{c,i}^\top \hat{\beta}_c. \quad (2)$$

Encoding, imputation, scaling and category handling were confined to the training fold, and a training-fold Breslow baseline hazard converted the linear predictor to 24-month risk.

Each usable optional modality was represented as an unordered token containing measurements, modality identity, usability/status indicators and quality features. A permutation-invariant Deep Sets residual encoder processed the available token set:

$$h_i = \rho \left(\frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \phi(x_{im}, q_{im}, e_m) \right), \quad \Delta\eta_{\text{res},i} = g(h_i). \quad (3)$$

The fused score was

$$\eta_{V1R,i} = \eta_{V0,i} + \lambda_f \Delta\eta_{\text{res},i}. \quad (4)$$

The backbone contained 3,225 parameters. When $\mathcal{M}_i$ was empty, both the residual and fused-score deviation were exactly zero, so the returned prediction equalled V0. This property made fallback deterministic rather than learned and allowed full-coverage performance to be evaluated without replacing unavailable patients.

4.5 Nested cross-fitting and optimization

Development used five outer folds, three inner folds and repetition seeds 17, 29, 43, 71 and 101. For each outer fold f , preprocessing, feature handling, model fitting, residual penalty, optimization checkpoint, shrinkage scale and baseline-hazard estimation were confined to training data. Inner cross-validation searched residual penalties $\{0.01, 0.1, 1.0\}$, checkpoints $\{0, 10, 25, 50\}$ and residual scales $\{0.1, 0.2, 0.3, 0.4, 0.5, 1.0\}$. No outer-fold outcome information was used to choose these values. Repeated predictions yielded 25 matched V0/V1R members (five seeds by five outer folds), allowing both fold-level pairing and seed-level direction summaries.

4.6 Development calibration bridge

For held-out outer fold f , raw V1R risk was transformed by

$$x_{\text{raw},i} = \text{logit}(p_{\text{V1R},i}), \quad (5)$$

$$x_{\text{bridge},i} = \alpha_{\text{logloss},f} + 0.90x_{\text{raw},i}, \quad p_{\text{bridge},i} = \sigma(x_{\text{bridge},i}). \quad (6)$$

The intercept was fitted by weighted log-loss using only non-held-out development folds. The slope was fixed positive, and no pattern-specific parameters were introduced. Consequently, the bridge was global, strictly monotone and rank-preserving, and it did not alter V0, raw V1R, fallback or coverage. The diagnostic population comprised 2,460 repeated out-of-fold rows and 415 evaluable events. Prespecified gates were global Brier change $\leq +0.0005$, worst supported-pattern regret $\leq +0.005$, improved calibration-in-the-large and slope, preserved ranking and full coverage.

4.7 Locked confirmation protocol

Before confirmation outcomes were accessed, the protocol, input manifest, dependency lock and patient-level prediction artifact were frozen. Twenty-five matched members were generated using development-fold fitting only, with no confirmation tuning, bridge, router, refitting, threshold selection or patient removal. Outcomes were read only after prediction sealing. The primary readout was raw V1R. Aggregate uncertainty used 2,000 patient-level stratified bootstrap replicates (seed 20260825). U5R8 separately applied the already frozen $\beta = 0.90$ bridge with the arithmetic mean development intercept $\alpha = -0.1450338$, using 2,000 event-stratified replicates (seed 20260827); it was labelled a locked secondary post-unseal readout.

4.8 Selective routing and extension analyses

The U6R1 router first averaged repeated out-of-fold predictions to one row per patient. A five-fold patient-level cross-fitted logistic router selected raw V1R or V0 using either core value features or reliability-augmented features, including across-repetition standard deviations of V0 risk, V1R risk and their difference. Fixed thresholds were 0.40, 0.50 and 0.60; 2,000 bootstrap replicates quantified uncertainty.

U7R2 evaluated adapted clinical/radiomics comparator families C1–C4 on the held-out RADCURE test split. U8 replicated residual shrinkage with transcriptome

tokens in TCGA-HNSC: 14,417 common genes were represented as within-sample ranks, each training fold retained the 500 most variable genes, and a Linear(500→32), Tanh, Linear(32→1) residual head supplied a transcriptome residual. U8E fitted one T1R model on the complete TCGA development cohort and applied it without cohort-specific refitting to GSE65858 and GSE41613. Detailed definitions and aggregate outputs are provided in Supplementary Information sections S8–S11.

4.9 Estimands and statistical analysis

The paired primary estimand was

$$\Delta\text{Brier}_{24} = \text{Brier}_{24}(\text{V1R}) - \text{Brier}_{24}(\text{V0}), \quad (7)$$

with analogous deltas for Uno C, horizon AUC and Harrell C. Lower Brier and higher discrimination values are better. Calibration-in-the-large is better closer to zero; calibration slope is better closer to one. Pattern safety used

$$\mathcal{R}_{\text{worst}} = \max_{p \in \mathcal{P}_{\text{supported}}} \{ \text{Brier}_{24}(\text{candidate}, p) - \text{Brier}_{24}(\text{reference}, p) \}. \quad (8)$$

IPCW Brier scores, Uno C and time-dependent AUC addressed censoring [16, 17]. Bootstrap resampling was performed at patient level, preserving the paired nature of model comparisons. Two-sided 95% intervals are percentile bootstrap intervals from the prespecified replicate sets.

4.10 Reproducibility, artifact governance and ethics

Repetition seeds, split roles, hash identities and aggregate-source paths were recorded for each analysis. The formal development out-of-fold SHA256 was E43BB6C0D8E2C7F9C8B22A9C416AD752109B926A8526273D88811458CD73BCE0; the confirmation protocol SHA256 was 1E134D30557D1CC153997F6EBB79E99C2E160F85A9F94BF3BBDD81C2C588BDED; and the sealed prediction artifact SHA256 was E5DA83BDB3BB97C3488687C78CCF166FBD03059619DD0690E2767CAF8F393FCF. Two V0 reruns produced identical out-of-fold hashes. All result figures and tables were generated from aggregate JSON artifacts; no patient-level predictions are included in the manuscript repository.

This analysis used retrospective datasets under their original governance and access conditions; no participants were newly recruited. Dataset-specific review-board approval, consent/waiver language, accessions, versions and secondary-analysis exemptions remain to be reproduced before submission. No patient or public involvement has yet been documented. A large-language-model-assisted coding and writing tool helped organize code, wording and LaTeX; it was not an author and did not independently generate or validate results. Final author verification is required.

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Data availability

HANCOCK, TCGA-HNSC, GSE65858, GSE41613 and RADCURE are available subject to source conditions [1, 3–6]. The repository tracks aggregate metrics, frozen specifications, audits and figure-generation code. Patient-level predictions remain outside version control and will not be redistributed contrary to source terms. **[TBD: Add final accession/version table and archival DOI.]**

Code availability

The configuration-driven `trust-hn` repository contains model code, frozen contracts, aggregate audits and tests. **[TBD: Add public archival release, licence, environment lock and runtime details.]**

Protocol and analysis availability

The frozen confirmation protocol, analysis specification, aggregate result manifests and artifact hashes are retained with the study materials. Patient-level prediction files remain access-controlled. **[TBD: Add the permanent protocol and analysis-plan archive or the process for reasonable-access requests.]**

Author contributions

[TBD: Complete CRediT roles for all authors.]

Competing interests

The authors declare **[TBD: no competing interests / specify]**.

Additional information

Supplementary information: the Supplementary Information provides sections S1–S16, Figs. S1–S4 and Tables S1–S8, covering evidence architecture, cohort and leakage procedures, model implementation, locked confirmation, reliability routing, RADCURE benchmarking, TCGA method replication and GEO platform transport. The completed TRIPOD+AI checklist and PROBAST+AI assessment will accompany the submission package.

Correspondence: **[TBD: corresponding author]**.

Provenance and peer review: **[TBD: journal completion]**.

Supplementary Information

This Supplementary Information expands the main-text Methods and Results. It documents the evidence architecture, cohort contracts, model implementation, leakage controls, statistical plan, locked confirmation, routing exploration and extension analyses in RADCURE, TCGA-HNSC and GEO. All numerical summaries are derived from frozen aggregate artifacts; no patient-level predictions are included.

S1 Evidence architecture and claim map

The study was organized as a staged evidence pipeline. Repeated nested cross-fitting in the HANCOCK development cohort quantified the incremental value of the V1R residual pathway over a clinical-pathological anchor. A monotone calibration bridge was then optimized and evaluated within development folds. Raw V1R predictions and all analysis dependencies were sealed before confirmation outcomes were accessed. Finally, router, RADCURE, transcriptome and GEO analyses examined selective use, contemporary benchmarking, method transfer and platform transport.

This architecture separates four claims. First, V1R showed a reproducible development gain, favourable in all five repetition seeds, with complete coverage and pattern-level safety within the prespecified boundary. Second, the HANCOCK confirmation cohort showed favourable point estimates for every principal metric; the bootstrap intervals reflected the moderate event count and supported a consistent directional signal. Third, the development bridge improved calibration summaries, whereas the later bridge readout identified cohort-level probability recalibration as the next step for transportable absolute-risk output. Fourth, router and extension analyses provide design insight and transport evidence complementary to the locked raw-V1R result.

This separation is a strength of the study. Model development, calibration, confirmation and extension are distinguished by frozen protocols and explicit artifact boundaries. The central finding—controlled incremental multimodal evidence can improve risk ranking while preserving full output availability—can therefore be evaluated alongside calibration, routing and transport considerations without conflating exploratory signals with locked confirmation.

S2 HANCOCK cohorts, prediction time and endpoint

HANCOCK was the primary development and confirmation ecosystem. The development analysis included 610 eligible postoperative patients with 173 deaths; the locked confirmation analysis included 152 patients with 40 events. Of 763 source records, one was excluded and 762 eligible postoperative patients remained. The split preserved a separately sealed outcome cohort while retaining sufficient development events for repeated nested cross-fitting.

For patient i , T_i denotes event time, C_i censoring time, $Y_i = \min(T_i, C_i)$ observed duration and $\delta_i = I(T_i \leq C_i)$ the event indicator. Follow-up began at first treatment and ended at last information; the event was all-cause death. Non-positive durations were excluded. The primary horizon was $t^* = 730.5$ days, approximately 24 months.

The intended prediction time was immediately after definitive surgery and pathological review, using information available at that point and excluding post-prediction outcome information.

The clinical anchor used age, sex, smoking, primary site, grade, p16, resection, pathological T category and pathological N category. Optional blood, ICD-coded comorbidity and tissue-microarray inputs were admitted only through the residual pathway. This ordering mirrors clinical deployment: the anchor contains routinely available postoperative variables, whereas optional modalities may differ in completeness and preprocessing success.

S3 Acquisition, usability and quality contract

For optional modality m , the contract distinguished acquisition a_{im} , usability u_{im} and an outcome-blind quality vector q_{im} . Acquisition indicated that a source record or assay existed. Usability indicated that the frozen preprocessing contract could produce a valid model input. The usable set was

$$\mathcal{M}_i = \{m : u_{im} = 1\}. \quad (9)$$

Quality summaries included missing fraction, out-of-range counts, assay completeness and distributional distance from training data. Absent and acquired-but-unusable measurements were represented structurally rather than as zero-valued biology. The natural acquisition pattern was the blood–ICD–TMA usability tuple. Pattern counts and quality distributions were computed without outcomes. For development safety analyses, a supported pattern required at least 30 patients and 10 events; rarer patterns were descriptive and did not receive pattern-specific tuning.

Of 762 eligible patients, 693 blood, 712 ICD and 736 tissue-microarray records were acquired, whereas 683, 712 and 736, respectively, were usable. Thus, ten acquired blood records failed the usability contract. Complete blood and TMA records numbered 604 and 639; 79 blood and 97 TMA records were partially usable. This contract preserves the distinction between absence, partial usability and acquired-but-unusable evidence.

The clinical-pathological anchor was fixed to age at diagnosis, sex, smoking status, primary tumour site, grade, p16 association, resection status, pathological T category and pathological N category. Pathology belonged to the postoperative anchor rather than being treated as an optional token. Adjuvant treatment, recurrence, progression, metastasis recorded after the prediction time and outcome-derived variables were prohibited predictors. Optional blocks remained separate at the data boundary; they were not concatenated into a complete-case matrix before availability and usability had been assigned.

The blood block used a fixed panel of 16 haematology measurements from the prespecified pre-treatment window. A missing blood row denoted non-acquisition, whereas a present row with invalid or entirely missing measurements could be acquired but unusable. Partially observed usable rows retained within-block missingness for fold-bound imputation and explicit missingness indicators. TMA followed the same structural rule: absence of a row represented modality absence; missing values

within an existing row represented within-modality incompleteness. This distinction prevented biological zeros from being substituted for unavailable assays.

The locally available ICD block consisted of 40 pre-extracted code groups. The underlying vocabulary had been generated with a corpus-level minimum document frequency of three, and the raw text source was not available for reconstruction. ICD was therefore marked as having conditional provenance. It was retained in the frozen evidence contract, but this limitation is disclosed because a fully fold-pure vocabulary sensitivity analysis would be preferable in a future release.

Separate fold-bound preprocessors handled the clinical mixed-type block and the three numeric optional blocks. Training-fold medians, scaling parameters, categorical levels and one-hot mappings were estimated only from explicitly allowed fit identifiers. New categories at transform time mapped to an unknown level. The preprocessing interface rejected held-out identifiers during fitting and returned missingness indicators alongside transformed values. These checks made leakage control executable rather than dependent only on analyst convention.

S4 Clinical anchor and residual network

The V0 anchor was an elastic-net Cox proportional-hazards model [8, 9]. After training-fold preprocessing,

$$\eta_{V0,i} = x_{c,i}^\top \hat{\beta}_c. \quad (10)$$

Encoding, imputation, scaling and category handling were confined to the corresponding training fold. A training-fold Breslow baseline-hazard estimator converted the linear predictor to 24-month mortality risk.

V1R was a clinically anchored residual Deep Sets Cox backbone with 3,225 parameters. Each usable modality was represented as an unordered token containing its measurements, learned representation, modality identity, usability status and quality indicators:

$$h_i = \rho \left(\frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \phi(x_{im}, q_{im}, e_m) \right), \quad \Delta\eta_{\text{res},i} = g(h_i). \quad (11)$$

The fused score was

$$\eta_{V1R,i} = \eta_{V0,i} + \lambda_f \Delta\eta_{\text{res},i}. \quad (12)$$

The token form makes the input set permutation invariant, while the masked mean allows any non-empty usable subset to be processed. When $\mathcal{M}_i$ is empty, the residual and fused-score deviations are exactly zero, so V1R returns V0 deterministically.

S5 Nested cross-fitting, optimization and leakage control

Repeated nested cross-fitting used five outer folds, three inner folds and repetition seeds 17, 29, 43, 71 and 101. Preprocessing, feature selection where applicable, model fitting, shrinkage selection and baseline-hazard estimation were confined

to training folds. For each outer fold f , inner cross-validation searched residual penalties $\{0.01, 0.1, 1.0\}$, optimization checkpoints $\{0, 10, 25, 50\}$ and residual scales $\{0.1, 0.2, 0.3, 0.4, 0.5, 1.0\}$.

This design yields 25 matched V0/V1R members (five seeds by five outer folds) and supports paired fold-level and seed-level summaries. Outer-fold outcomes were not used for hyperparameter selection. Repeated out-of-fold rows provided the development performance estimates and the calibration-bridge diagnostic, while seed-level paired differences assessed reproducibility. No confirmation outcomes entered development tuning.

The V1R stage was a transparently result-informed rescue of the original unscaled V1 analysis. The original U2 decision, **V1_DOES_NOT_EARN_COMPLEXITY**, remains unchanged. Prediction-space sensitivity analysis indicated that the unscaled residual was too aggressive; scales near 0.2–0.4 reduced pattern regret and calibration-slope deterioration while retaining a small, seed-stable discrimination signal. V1R therefore added residual scale to the inner-loop search and was labelled post-hoc exploratory rather than retrospectively reclassifying the original V1 result.

The frozen V1R selection objective minimized mean inner-fold IPCW Brier score at 730.5 days, with mean inner-fold Uno C as the secondary criterion. Deterministic ties favoured the larger residual penalty, then the smaller residual scale, then fewer optimization steps. Architecture search and broad hyperparameter sweeps were prohibited. The implementation used deterministic PyTorch computation on CPU in float64, Adam optimization with learning rate 0.003, weight decay 0.0001 and gradient clipping at norm 5.0. The loss combined mean negative Breslow partial likelihood with an L2 penalty on the residual score.

After the inner-loop candidate was selected, the corresponding outer-training fold was refitted and its Breslow baseline hazard was estimated after residual scaling using training outcomes only. Each patient was predicted exactly once per repetition. Patient-level out-of-fold predictions were stored outside tracked manuscript artifacts; only aggregate audits and frozen specifications were committed. The rescue gate retained the original safety thresholds and could authorize only a development signal, not confirmatory superiority or external validation.

S6 Global calibration bridge

The U5R7 bridge was frozen as **beta_0.900_logloss**. For raw V1R 24-month risk p_{V1R} in held-out outer fold f ,

$$x_{\text{raw},i} = \text{logit}(p_{V1R,i}), \quad (13)$$

$$x_{\text{bridge},i} = \alpha_{\text{logloss},f} + 0.90x_{\text{raw},i}, \quad p_{\text{bridge},i} = \sigma(x_{\text{bridge},i}). \quad (14)$$

The intercept was fitted by weighted log-loss using only non-held-out development folds. The bridge was global and pattern-agnostic, with a fixed positive slope. It was therefore strictly monotone, preserved ranking, retained 100% coverage and did not alter the exact V0 fallback.

The diagnostic population contained 2,460 repeated out-of-fold rows and 415 evaluable events. Gates were global Brier change $\leq +0.0005$, worst supported-pattern regret $\leq +0.005$, improved calibration-in-the-large and calibration slope, ranking preservation and full coverage. Confirmation outcomes were not used for fitting or tuning.

U5R7 followed a sequence of development-only bridge explorations and froze the candidate identifier `beta_0.900_logloss` before its diagnostic readout. The positive slope was not estimated separately by pattern, and the intercept was fitted anew within each non-held-out development partition. Thus, the bridge could move the absolute probability scale without changing within-fold ordering. It also left the raw V1R score, V0 anchor, acquisition rules and empty-set fallback untouched.

The bridge diagnostic is not numerically interchangeable with the primary development table. It contains repeated evaluable out-of-fold rows and addresses whether a global monotone transformation improves absolute-risk calibration; the primary table compares V0 and raw V1R as prediction models. The small positive Brier change was retained in the report rather than hidden. U5R7 passed its prespecified development-safe gates, but the decision authorized only a development calibration candidate. Raw V1R remained the primary prediction path and the only fused model carried into the locked confirmation comparison.

S7 Locked confirmation and secondary bridge protocol

The confirmation protocol, input manifest, dependency lock and prediction artifact were frozen before confirmation outcomes were unsealed. Twenty-five matched V0/V1R members were generated using five seeds and five outer folds, with development-fold fitting only and no confirmation tuning, bridge, router, refitting or selective patient removal. Outcomes were read after prediction sealing.

Aggregate uncertainty used 2,000 patient-level stratified bootstrap replicates (seed 20260825). The primary confirmation readout was raw V1R. U5R8 separately reused the frozen $\beta = 0.90$ bridge with the arithmetic mean development intercept $\alpha = -0.1450338$, using 2,000 event-stratified replicates (seed 20260827). Because U5R8 occurred after raw-outcome unsealing, it is reported as a locked secondary post-unseal validation and is interpreted as evidence about probability-scale transport rather than ranking.

The locking sequence separated prediction generation from outcome evaluation. The sealed input manifest identified the eligible confirmation records and the exact source variables consumed by V0 and V1R. The dependency receipt fixed the execution environment, and the prediction receipt recorded the patient-level artifact before the event file was joined. Aggregate metrics were then computed from the sealed predictions. The confirmation cohort therefore supplied no information for preprocessing, hyperparameter choice, residual shrinkage, baseline-hazard estimation, threshold selection or model refitting.

The 25 members were kept matched between V0 and V1R so that every paired contrast compared predictions obtained under the same repetition and outer-fold training role. Coverage was evaluated before performance summaries and remained 100% for

both models. Bootstrap sampling operated on patients rather than the 25 repeated prediction rows; this preserved within-patient dependence and prevented artificial precision from treating repeated predictions as independent observations. The bridge was excluded from the primary confirmation estimand because it had been developed as a separate calibration layer. Its later reuse in U5R8 did not modify the locked raw-V1R result.

S8 Reliability-aware router design

The U6R1 router evaluated whether patient-level reliability information could choose between raw V1R and V0. Repeated out-of-fold predictions were first averaged to one row per patient. A five-fold patient-level cross-fitted logistic router used either core value features or reliability-augmented features. Reliability features included across-repetition standard deviations of V0 risk, V1R risk and their difference. Thresholds 0.40, 0.50 and 0.60 were fixed before readout.

Router performance was compared with both V0 and direct raw V1R. The primary probabilistic readout was IPCW Brier score at 24 months. FUSE rate, delta versus V0, delta versus raw V1R and rank concordance with raw V1R summarized behaviour. Bootstrap resampling used 2,000 replicates. This matched-coverage design tests whether stability information adds routing value without rewarding patient exclusion.

The router used fixed logistic regularization $C = 0.5$. A FUSE decision returned raw V1R; a FALLBACK decision returned V0. It did not abstain, remove patients or create a third risk model, so every policy retained 100% prediction availability. Core features summarized the cross-fitted value contrast between the two candidate predictions. The reliability-augmented version added the across-repetition standard deviation of V0 risk, V1R risk and their patient-level difference, thereby asking whether a stable apparent gain should be treated differently from an equally large but unstable gain.

The thresholds were intentionally reported as a panel rather than selected after observing the final metric. At lower thresholds, more patients followed V1R; at higher thresholds, the router required stronger evidence before fusing. Rank concordance with raw V1R was included because switching individual patients back to V0 can change global ordering even when Brier score improves. Direct raw V1R remained the strongest point-estimate policy, and the best router’s bootstrap interval crossed zero. U6R1 therefore supports reliability-aware selection as a research question, not as a validated deployment component or a replacement for the full-coverage primary path.

S9 RADCURE comparator characterization design

U7R2 evaluated adapted clinical/radiomics comparator families C1–C4 in the RAD-CURE held-out test split. The analysis included 626 patients and 110 events. The purpose was to benchmark a contemporary clinical/tumour-volume modelling task and to characterize discrimination, probabilistic error and calibration in a held-out imaging-rich cohort. Comparators were adapted to the RADCURE evidence contract; the current HANCOCK blood–ICD–TMA V1R artifact was not evaluated because the

frozen optional-modality inputs and preprocessing contract did not map directly onto RADCURE.

The characterization reports IPCW Brier score, Uno C at 24 months, time-dependent AUC, Harrell C, calibration-in-the-large and calibration slope. This analysis provides a strong contextual benchmark and shows the practical value of combining clinical predictors with tumour-volume information. It is interpreted as comparator characterization rather than current-V1R external validation.

Before U7R2, the external-cohort intake procedure was frozen for a possible formal current-V1R confirmation. A qualifying cohort had to reproduce the required postoperative prediction time, clinical-pathological anchor variables, blood, ICD and TMA evidence definitions, endpoint, follow-up horizon and outcome-untouched status. The available inventory contained no cohort satisfying all of these conditions. RADCURE was therefore chosen only for the narrower task that its clinical and radiomic variables could support.

The held-out RADCURE test split was not used to tune any HANCOCK model, calibration bridge, router threshold or safety gate. Conversely, its comparator families were adapted to the RADCURE data contract and are not numerical surrogates for the current V1R artifact. RADCURE outcomes had also been consumed in earlier work, precluding an outcome-untouched confirmation label. These constraints are why the analysis is described as external characterization: it tests the plausibility of combining clinical information with an image-derived tumour-volume representation, but does not establish transportability of blood-ICD-TMA residual fusion.

S10 TCGA transcriptome residual replication design

U8 tested transfer of the residual-shrinkage principle from sparse HANCOCK optional modalities to a dense transcriptome modality. The analysis included 519 TCGA-HNSC patients and 221 events. The clinical anchor used age, sex, site, stage, HPV status, treatment and smoking. The transcriptome contained 14,417 common genes represented as within-sample ranks; each training fold retained the 500 most variable genes.

The residual head was Linear(500→32), Tanh, Linear(32→1), and the fused score was

$$\eta_{\text{T1R},i} = \eta_{\text{V0},i} + \lambda_f \Delta \eta_{\text{transcriptome},i}. \quad (15)$$

All selection steps were confined to training folds. The completed run produced 2,595 out-of-fold rows with complete coverage and an exact fallback error of zero. Because TCGA outcomes had already been consumed during analysis development, this experiment is labelled post-hoc method replication: it tests whether the architectural principle transfers to transcriptome data, rather than providing a pristine external validation of V1R.

The replication used five outer folds, three inner folds and the same repetition seeds 17, 29, 43, 71 and 101. Within every outer-training partition, clinical preprocessing, elastic-net anchor selection, the 500-gene variance filter, residual fitting, residual-penalty choice, optimization checkpoint, shrinkage scale and Breslow baseline hazard were estimated without the held-out fold. The transcriptome residual head

contained 16,065 trainable parameters, below the frozen 50,000-parameter ceiling. The acquisition-pattern safety check collapsed to the single transcriptome-present pattern, while the score construction retained exact zero-residual fallback as a structural test.

The frozen exploratory gate required complete coverage, structural fallback safety, bounded Brier deterioration, bounded calibration deterioration and at least one reproducible incremental-value path. Observed worst supported-pattern Brier regret was +0.002714 against a +0.020 ceiling. Mean absolute calibration-in-the-large deterioration was +0.049232 against a +0.100 ceiling, and mean absolute calibration-slope-error deterioration was -0.084226 against a +0.150 ceiling. Both probability-error and discrimination paths met their seed-stability requirements, so the frozen gate returned `T1R_EARNS_COMPLEXITY`.

That gate result has a deliberately limited meaning. It authorizes the statement that the residual-shrinkage principle showed internal value in a dense transcriptome setting; it does not establish current-V1R external validity, pristine confirmation, clinical utility or deployment readiness. It also does not replace the HANCOCK backbone or alter the locked raw-V1R confirmation. Patient-level TCGA predictions remain outside the manuscript repository, and the publication table is generated from the aggregate development audit.

S11 GEO cross-platform transfer design

U8E applied a single full-development T1R fit to GSE65858 and GSE41613 without cohort-specific refitting, tuning or threshold selection. GSE65858 contributed 244 patients and 78 events; GSE41613 contributed 97 patients and 51 events. Before application, inner cross-validation selected an elastic-net anchor with alpha 0.05 and L1 ratio 0.1, residual penalty 1.0, optimization checkpoint 10 and residual scale 1.0.

Within-sample rank representation was retained as the transcriptome input representation. The analysis reports paired Brier, Harrell C, Uno C, time-dependent AUC, calibration-in-the-large and calibration-slope changes. The design emphasizes method transfer and cross-platform signal retention. Calibration slopes above one indicate compressed predicted-risk dispersion, making cohort-level recalibration the natural next step before prospective absolute-risk reporting.

The full-development TCGA fit contained the same 16,065-parameter residual head used during cross-fitting. Development-only inner cross-validation selected elastic-net $\alpha = 0.05$, L1 ratio 0.1, residual penalty 1.0, optimization checkpoint 10 and residual scale 1.0 before either GEO cohort was evaluated. The resulting feature-selection and model parameters were carried forward as one frozen package. No GEO-specific gene selection, refitting, hyperparameter search, decision threshold or outcome-dependent cohort selection was performed, and coverage was complete in both cohorts.

GSE65858 provided a conventional test of whether the transcriptome residual preserved ordering beyond a varying clinical anchor. GSE41613 provided a complementary edge case: the applied anchor had no cross-patient variation, so its discrimination was 0.5 and any recovered ordering arose from the transported transcriptome component. In both cohorts, Brier score and all discrimination summaries changed favourably, while calibration behaviour remained cohort dependent. This combination of results

distinguishes transport of relative ordering from transport of absolute probability scale.

Because both GEO outcome sets had contributed to earlier Phase 6 analyses, the application is explicitly post-hoc characterization. Concordant directions across the two cohorts strengthen the plausibility of the transcriptome residual signal, but they are not treated as formal external validation. The appropriate next experiment is to seal predictions in a newly identified outcome-untouched cohort, prespecify any cohort-level recalibration, and evaluate both discrimination and calibration without reusing these GEO outcomes for tuning.

S12 Implementation and statistical plan

Paired development and confirmation estimands were

$$\Delta\text{Brier}_{24} = \text{Brier}_{24}(\text{V1R}) - \text{Brier}_{24}(\text{V0}) \quad (16)$$

and analogous discrimination deltas. Lower Brier and higher discrimination values are better. Calibration-in-the-large is better closer to zero; calibration slope is better closer to one. Pattern safety used

$$\mathcal{R}_{\text{worst}} = \max_{p \in \mathcal{P}_{\text{supported}}} \{\text{Brier}_{24}(\text{candidate}, p) - \text{Brier}_{24}(\text{reference}, p)\}, \quad (17)$$

where supported patterns required at least 30 patients and 10 events.

IPCW Brier scores, Uno concordance and horizon AUC addressed censoring [16, 17]. Two-sided 95% percentile intervals used prespecified patient-level stratified bootstrap replicate sets. Patients, rather than repeated prediction rows, were resampled for confirmation intervals. Development summaries reported both aggregate means and the number of repetition seeds with a favourable direction. No multiplicity adjustment was applied to exploratory extension readouts; their role is supportive characterization.

The primary horizon was 730.5 days. IPCW weights were estimated within the relevant analysis population, and paired comparisons used the same eligible patients and horizon for both candidate predictions. Uno concordance summarized censoring-adjusted ranking through the horizon; time-dependent AUC summarized case-control separation at the horizon; Harrell concordance was retained as a familiar global ranking measure. IPCW Brier score combined discrimination and calibration as a proper probabilistic loss. Calibration-in-the-large compared average predicted and observed risk on the fitted probability scale, while the calibration slope assessed dispersion.

For natural evidence-pattern analyses, a pattern entered the supported set only when at least 30 patients and 10 events were available. This threshold prevented small cells from becoming independently tuned strata. Rare patterns remained visible in counts and descriptive outputs, but they were not used to justify pattern-specific transport claims. Full-coverage performance always retained patients with no usable optional modality, whose deterministic V0 fallback made the deployment estimand different from a complete-case analysis.

Development cross-fitting produced repeated predictions for stability assessment, but uncertainty calculations respected the patient as the sampling unit. Confirmation intervals were computed from 2,000 patient-level stratified replicates. Router intervals likewise used 2,000 patient bootstrap replicates after repeated predictions had first been aggregated to one row per patient. Extension analyses report prespecified aggregate and seed-direction summaries without multiplicity-adjusted inferential claims. This hierarchy keeps the locked confirmation, development diagnostics and post-hoc characterizations statistically distinct.

S13 Supporting development and calibration evidence

V1R improved every aggregate discrimination measure and the mean 24-month IPCW Brier score in HANCOCK development (Table S1). The mean Uno C gain was +0.021303 and was favourable in all five seeds. Coverage remained 100%, the empty-set fallback error was zero, and the worst supported-pattern Brier regret was +0.009848, below the +0.020 safety boundary.

The U5R7 bridge moved calibration-in-the-large from +0.011680 to −0.003796 and calibration slope from 0.825887 to 0.881552 (Table S2). Ranking and coverage were preserved. The global Brier change was +0.000147, within the +0.0005 gate, and worst supported-pattern regret was +0.004176, below the +0.005 gate. Thus, the development bridge improved the probability scale without changing patient ordering or availability.

Table S1: Development performance of V1R

Metric	Clinical anchor	V1R	Difference	Interpretation
IPCW Brier at 24 months	0.124745	0.124338	−0.000407	Lower probabilistic error
Uno C at 24 months	0.644239	0.665542	+0.021303	Improved discrimination
Time-dependent AUC at 24 months	0.662022	0.682511	+0.020488	Improved horizon ranking
Harrell C	0.623021	0.633710	+0.010690	Improved overall ranking
Coverage	100%	100%	0	Full output availability
Empty-set fallback error	0	0	0	Exact clinical fallback
Worst supported-pattern Brier regret	Reference	+0.009848	+0.009848	Below +0.020 boundary
Seeds with improved Uno C	–	5/5	–	Consistent direction

Table S2: U5R7 development calibration bridge

Metric	Raw V1R	U5R7 bridge	Change	Interpretation
IPCW Brier at 24 months	0.126637	0.126784	+0.000147	Within +0.0005 gate
Calibration-in-the-large	+0.011680	−0.003796	−0.015477	Absolute error improved by 0.007884

Table S2 continued from previous page

Metric	Raw V1R	U5R7 bridge	Change	Interpretation
Calibration slope	0.825887	0.881552	+0.055665	Absolute error improved by 0.055665
Worst supported-pattern Brier regret	Reference	+0.004176	+0.004176	Below +0.005 gate
Coverage	100%	100%	0	Preserved
Ranking	Reference	Preserved	–	All held-out folds

S14 Locked confirmation and routing readouts

In the 152-patient, 40-event outcome-sealed cohort, raw V1R showed favourable point estimates for discrimination, Brier score and calibration summaries (Table S3). Patient-level stratified bootstrap with 2,000 replicates gave a mean delta Uno C of +0.021654 (95% interval -0.025965 to $+0.070816$) and a mean delta IPCW Brier of -0.005222 (95% interval -0.012843 to $+0.002171$). The direction was favourable across the principal readouts, with intervals reflecting the moderate event count.

The secondary U5R8 bridge readout preserved ranking and coverage but changed IPCW Brier from 0.134697 to 0.135782, calibration-in-the-large from +0.156751 to +0.160944 and calibration slope from 1.510335 to 1.678150. This result identifies cohort-level probability recalibration as the key next step for transportable absolute-risk output.

Table S3: Locked raw-V1R confirmation

Metric at 24 months	Clinical anchor	Raw V1R	Difference	Interpretation
Uno C	0.782579	0.804242	+0.021664	Favourable point estimate
IPCW Brier	0.136412	0.131246	-0.005166	Favourable point estimate
Time-dependent AUC	0.801094	0.813637	+0.012543	Favourable point estimate
Harrell C	0.733253	0.752842	+0.019589	Favourable point estimate
Calibration-in-the-large	0.211270	0.116482	-0.094788	Closer to zero
Calibration slope	1.945126	1.501223	-0.443903	Closer to one
Coverage	100%	100%	0	Preserved

Every router improved on the clinical anchor at the point estimate, while direct raw V1R remained the strongest overall policy (Table S4 and Fig. S1). The best router used reliability features at threshold 0.60: delta Brier versus V0 was -0.000506 , and its bootstrap interval was approximately -0.00162 to $+0.00076$. These results show that repetition-level stability contains useful routing information and support further development of selective prediction at matched coverage.

Table S4: Development-only router policies

Policy	FUSE rate	Brier ₂₄	Δ vs V0	Δ vs raw V1R	Rank concordance
V0	0.0%	0.123927	0	+0.001162	0.867
Raw V1R	100.0%	0.122765	-0.001162	0	1.000

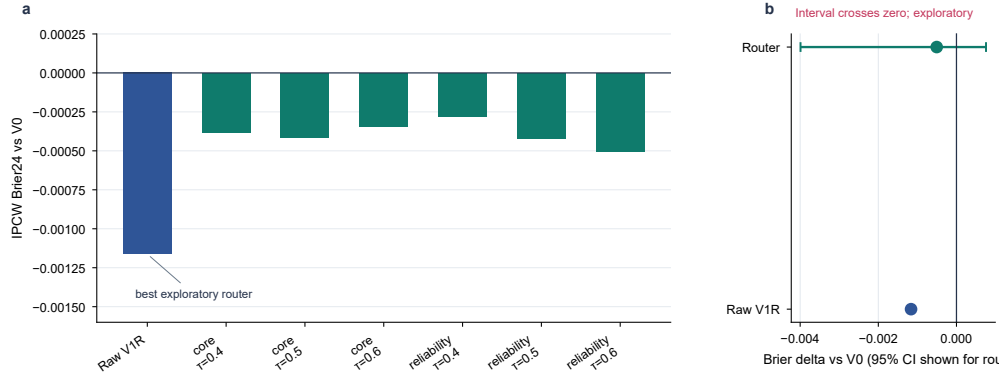


Fig. S1 Exploratory patient-level reliability router. **a**, IPCW Brier differences versus V0 for raw V1R and router policies. **b**, Best router point estimate with bootstrap interval. Reliability features improve routing relative to the core router and V0, while direct raw V1R remains the strongest full-coverage policy.

Table S4 continued from previous page

Policy	FUSE rate	Brier ₂₄	Δ vs V0	Δ vs raw V1R	Rank concordance
Core router, $q \geq 0.40$	66.1%	0.123545	-0.000382	+0.000780	0.945
Core router, $q \geq 0.50$	54.1%	0.123511	-0.000416	+0.000746	0.939
Core router, $q \geq 0.60$	40.3%	0.123585	-0.000341	+0.000821	0.928
Reliability router, $q \geq 0.40$	67.0%	0.123648	-0.000279	+0.000883	0.946
Reliability router, $q \geq 0.50$	54.8%	0.123508	-0.000419	+0.000743	0.938
Reliability router, $q \geq 0.60$	42.6%	0.123421	-0.000506	+0.000657	0.929

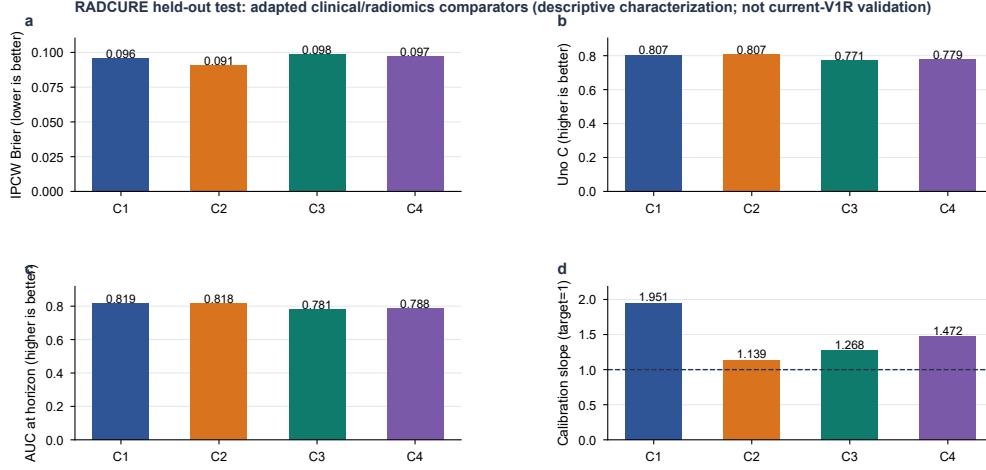


Fig. S2 RADCURE characterization of adapted clinical/radiomics comparators. a–d, IPCW Brier, Uno C, horizon AUC and calibration slope for C1–C4 in the held-out test split. C2 provides a strong benchmark by combining clinical predictors with tumour-volume information.

S15 Extension readouts: RADCURE, TCGA and GEO

The RADCURE held-out analysis characterized adapted clinical/radiomics comparator families in 626 patients with 110 events. The strongest comparator, C2, achieved Uno C 0.806744, 24-month AUC 0.818182 and IPCW Brier 0.090683 (Table S5 and Fig. S2). This analysis provides a strong contemporary benchmark for the multimodal prognostic task and highlights the value of combining clinical and tumour-volume information.

Table S5: RADCURE held-out comparator characterization

Comparator	Brier ₂₄	Uno C	AUC ₂₄	Harrell C	CITL	Calibration slope
C1	0.095801	0.806595	0.819376	0.795924	−0.097110	1.950857
C2	0.090683	0.806744	0.818182	0.797862	−0.044444	1.139466
C3	0.098469	0.771260	0.780742	0.761632	−0.099603	1.268349
C4	0.097403	0.779243	0.787882	0.768812	−0.074520	1.471544

T1R improved every discrimination measure and the mean Brier score in TCGA-HNSC (Table S6); discrimination improved in all five seeds (Table S7 and Fig. S3). The frozen gate returned `T1R_EARNS_COMPLEXITY`. This demonstrates that the clinically anchored residual-shrinkage principle transfers to a dense transcriptome modality.

Table S6: T1R development performance in TCGA-HNSC

Metric	Clinical anchor	T1R	Difference	Favourable seeds
IPCW Brier at 24 months	0.221795	0.218058	−0.003737	4/5

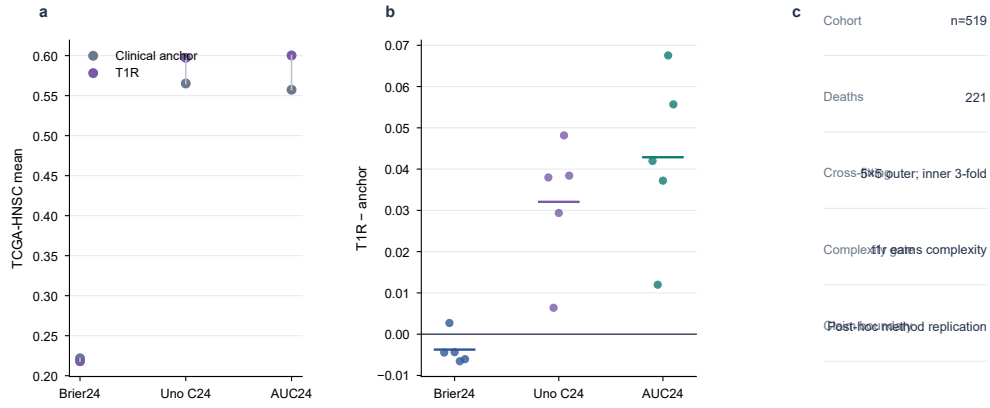


Fig. S3 T1R transcriptome method replication in TCGA-HNSC. **a**, Mean anchor and T1R performance. **b**, Seed-level T1R–anchor differences. **c**, Cohort role and claim label. Discrimination improved in all five seeds.

Table S6 continued from previous page

Metric	Clinical anchor	T1R	Difference	Favourable seeds
Harrell C	0.577281	0.605919	+0.028637	5/5
Uno C at 24 months	0.565199	0.597265	+0.032066	5/5
Time-dependent AUC at 24 months	0.557362	0.600236	+0.042874	5/5

Table S7: T1R paired changes by repetition seed

Repetition seed	Δ IPCW Brier ₂₄	Δ Uno C ₂₄
17	−0.004443	+0.037975
29	+0.002714	+0.006398
43	−0.004353	+0.029370
71	−0.006560	+0.048165
101	−0.006045	+0.038420

A single T1R model fitted to the complete TCGA-HNSC development cohort showed favourable Brier and discrimination changes in both GEO cohorts (Table S8 and Fig. S4). In GSE65858, Uno C increased from 0.583694 to 0.645818. In GSE41613, the applied clinical anchor had no cross-patient variation, whereas T1R restored patient-level ordering, with Harrell C 0.626361 and 24-month AUC 0.634011. Calibration slopes above one indicate compressed predicted-risk dispersion, making cohort-level recalibration a constructive next step for prospective absolute-risk reporting.

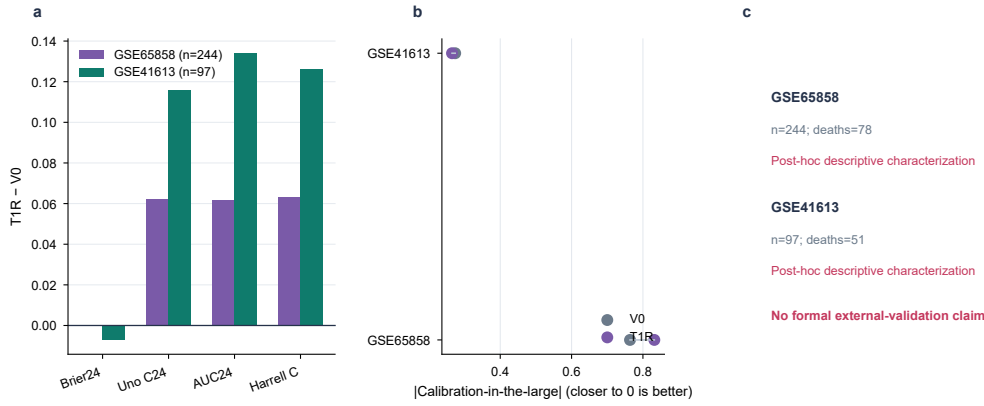


Fig. S4 GEO cross-platform characterization of T1R. **a**, T1R–anchor differences in GSE65858 and GSE41613. **b**, Absolute calibration-in-the-large error. **c**, Cohort sizes and evidence role. A single frozen T1R fit retained favourable discrimination changes across platforms.

Table S8: GEO transcriptome characterization

Cohort and metric	Clinical anchor	T1R	Difference
GSE65858 (n=244; 78 events)			
IPCW Brier at 24 months	0.195207	0.195070	−0.000137
Harrell C	0.581806	0.644868	+0.063062
Uno C at 24 months	0.583694	0.645818	+0.062123
Time-dependent AUC at 24 months	0.588972	0.650733	+0.061762
Calibration-in-the-large	−0.763793	−0.831959	−0.068166
Calibration slope	1.707442	1.941562	+0.234120
GSE41613 (n=97; 51 events)			
IPCW Brier at 24 months	0.265712	0.258607	−0.007104
Harrell C	0.500000	0.626361	+0.126361
Uno C at 24 months	0.500000	0.616039	+0.116039
Time-dependent AUC at 24 months	0.500000	0.634011	+0.134011
Calibration-in-the-large	−0.273717	−0.264792	+0.008925
Calibration slope	—	1.939018	—

S16 Reproducibility and artifact governance

The formal V1R development out-of-fold SHA256 was E43BB6C0D8E2C7F9C8B22A9C416AD752109B926A8526273D88811458CD73BCE0. The confirmation protocol SHA256 was 1E134D30557D1CC153997F6EBB79E99C2E160F85A9F94BF3BBDD81C2C588BDED, and the locked prediction artifact SHA256 was E5DA83BDB3BB97C3488687C78CCF166FBD03059619DD0690E2767CAF8F393FCF. Two V0 reruns produced identical out-of-fold hashes.

Aggregate sources included U5R7_V1R_beta09_logloss_bridge_candidate/selected_bridge_aggregate_results.json, U6R1_patient_level_router_

aggregation_exploration/u6r1_aggregate_results.json, U7R2_RADCURE_external_characterization/u7r2_radcure_external_characterization_aggregate_results.json, U8_T1R_transcriptome_residual_shrinkage/aggregate_t1r_transcriptome_development_cv_audit.json and U8E_T1R_external_characterization/aggregate_t1r_external_characterization_audit.json, all under research_studies/01_pattern_surv_hn/core_backbone/. Figure provenance is recorded in figures/figure_manifest.json; publication figures are generated by generate_figures.py.

Cohorts, preprocessing and stage definitions were configuration controlled. Repetition seeds were 17, 29, 43, 71 and 101. Each stage recorded analysis label, prior outcome access, tuning status and artifact integrity. Patient identifiers and patient-level predictions were excluded from version control; publication-facing tables and figures were generated from aggregate metrics, manifests and audits. This governance allows every central number to be traced to a frozen artifact without redistributing restricted patient-level predictions.

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